

HARSHWARDHAN PAYGUDE

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2026	B.Tech in Computer Science (AI&DS)	MIT World Peace University, Pune	7.40/10
2022	HSC (State Board)	B.H.Chate Jr Clg, Pune	89%
2020	CBSE (State Board)	Sinhgad spring dale shcool , Pune	71.00%

SKILLS AND EXPERTISE

- **Languages:** C, C++, Python, SQL
- **Frameworks:** TensorFlow, Keras, PyTorch, NumPy, Pandas, OpenCV
- **Databases:** MongoDB, SQLite, SQL
- **Tools and Platforms:** GitHub, Power BI, Tableau, Figma, PySpark
- **Generative AI:** LangChain, RAG Pipelines, Prompt Engineering, LLM Workflows, OpenAI APIs, Vector Databases, AI Agents, CNNs

WORK EXPERIENCE

- XR / Unity Development Intern - FireBirdVR** May'26
- Developed interactive educational VR experiences in Unity using C#, including immersive learning and narrative interactions.
 - Designed UI/UX workflows in Figma and collaborated in Agile sprint planning, testing, and iterative development.

PROJECTS

- MNIST Digit Classifier** April '25
- TensorFlow, Keras, NumPy, Python*
- Built a deep neural network achieving high digit classification accuracy on the MNIST dataset.
 - Tuned hyperparameters, applied ReLU and Softmax for robust model performance.
 - Visualized confusion matrices and learning curves for training insights.

- Car Sales Dashboard using Power BI** Jan '25
- Power BI, DAX, Excel*
- Developed an interactive dashboard to analyze car sales data across regions, brands, and time periods. Incorporated key visualizations such as bar charts (monthly sales by brand), pie charts (market share distribution), line charts (sales trends over time), and map visuals (regional sales performance).
 - Dynamic filtering and KPI tracking enabled to support data-driven business decisions.

- Vastu Interior Advisor using RAG-Augmented Agentic AI** Nov'25
- LangChain, Python, RAG, Generative AI, Vector Database, Telegram Bot*
- Developed a RAG-based AI interior advisor using LangChain for Vastu recommendations from room images.
 - Integrated object detection, embeddings, and vector database retrieval for context-aware suggestions.
 - Deployed the agentic AI workflow as a Telegram bot for real-time image analysis and redesign guidance.

- Mechanical Gear Defect Detection Using Image Processing** Dec'25
- CNN, SQLite3, Tkinter, ML*
- Developing an AI-assisted gear inspection system to detect faults using CNN-based image analysis.
 - Implemented user authentication, image upload, preprocessing (grayscale, resizing, normalization) & SQLite database.
 - Next phase: integrate CNN classification with AI-driven maintenance suggestions (replace gear, schedule service, etc.)..

RESEARCH PUBLISHED

- Published a research paper on CNN-based automated mechanical gear defect diagnosis at the IEEE International Conference for Convergence in Computing Technology (I3CTCON) (May 2026).
- Developed a deep learning predictive maintenance system using TensorFlow and Keras, achieving 98% accuracy in classifying six gear defect categories.

CERTIFICATIONS

- Unity Essentials Pathway – Unity Technologies
- IBM SkillsBuild – Data Fundamentals
- NAIP on ML & DL *by NIT Goa*
- GenAI in action *by IBM SkillsBuild*